Example 3

$S = \{HHH, HHT, HTH, HTT,$
$THH, THT, TTH, TTT\}$

$x \in \{0, 1, 2, 3\}$

$P(x=0) = 1/8$

$P(x=1) \quad 3/8$

$P(x=2) \quad 3/8$

$P(x=3 \quad 1/8$

$= P(Y=0) + P(Y=2) + \ldots$

$= 5/8 + 2/8 + 1/8$

$= 8/8 = 1$

PDF

$$F(x) = \begin{cases} 0 & x < 0 \\ 1/8 & 0 \le x < 1 \\ 4/8 \left(\frac{1}{8} + \frac{3}{8}\right) & 1 \le x < 2 \\ 7/8 & 2 \le x < 3 \\ 1 & x \geq 3 \end{cases}$$

$= \{HH, \ldots$

$P(X \le 1) = P(H \ldots$

Pool of 3 white 2 red balls. Two balls are taken at random. X = random variable the no. of white balls picked. describe

$n(S) = {}^5C_2 = 10$

$P(X=2) = 3/10$

$P(X=1) = 6/10$

$P(X=0) = 1/10$

$w_1w_2w_3 = w_1w_2 \; w_1w_3 \; w_2w_3 \; w_1r_1 \; w_2r_1 \; w_3r_1 \; w_1r_2 \; w_2r_2 \; w_3r_2 \; r_1r_2$

$X =$ Der...

$n(S)$

$P(X$

$P($

$P($

P

Eg. 3 Tossing a coin 3 times

X = no. of heads.

$S = \{HHH, HTH, THH, TTH, HHT, HTT, THT, TTT\}$

$n(S) = 8$

$P(X=3) = 1/8$

$P(X=2) = 3/8$

Eg 4

HHH TTT.
HHT TTH
HTT THT
HTH THH.

Example 6,

Y — no of consecutive tails.

i) $P(Y=0) = 5/8$

$P(Y=2) = 2/8$

$P(Y=3) = 1/8$

BW required which contains 90% of signal power

ie $\frac{90}{100} \times \frac{1}{2a} = \frac{0.9}{2a}$

$$\text{Power} = R_{xx}(0) = \int_{-\infty}^{\infty} H_{xx}(f)\,df \qquad H_{xx}(f) = \frac{1}{a^2 + 4\pi^2 f^2}$$

We cannot integrate over the entire frequency band since it decays to zero as f approaches ∞ or $-\infty$. So then we specify a certain $-W$ to W. That is what we are going to find to get the BW that contains 90% of the signal power only.

$$\Rightarrow \frac{0.9}{2a} = \int_{-W}^{W} \frac{1}{a^2 + 4\pi^2 f^2}\,df$$

Using the integral identity: $\boxed{\int \frac{1}{b^2 + u^2}\,du = \frac{1}{b}\tan^{-1}\left(\frac{u}{b}\right) + c}$

(u is alone so f should be alone)

$$\frac{0.9}{2a} = \int_{-W}^{W} \frac{1/4\pi^2}{\frac{a^2}{4\pi^2} + f^2}\,df = \frac{1}{4\pi^2}\int_{-W}^{W} \frac{1}{\left(\frac{a}{2\pi}\right)^2 + f^2}\,df \qquad \boxed{\begin{array}{l} u = f \\ b = a/2\pi \end{array}}$$

$$\frac{0.9}{2a} = \frac{1}{4\pi^2}\left[\frac{1}{a/2\pi}\tan^{-1}\left(\frac{f}{a/2\pi}\right)\right]_{-W}^{W}$$

$$\frac{0.9}{2a} = \frac{\cancel{2}\pi}{{}_2\cancel{4}\pi^{\cancel{2}} a}\left[\tan^{-1}\left(\frac{W}{a/2\pi}\right) - \tan^{-1}\left(\frac{-W}{a/2\pi}\right)\right] = \frac{1}{2\pi a}\left[\tan^{-1}\left(\frac{W}{a/2\pi}\right) \overset{--}{+} \tan^{-1}\left(\frac{W}{a/2\pi}\right)\right]$$

$$\frac{0.9}{\cancel{2}a} = \frac{1}{\cancel{2}\pi\cancel{a}} \cdot \cancel{2}\tan^{-1}\left(\frac{2\pi W}{a}\right)$$

$$\tan^{-1}\left(\frac{2\pi W}{a}\right) = \frac{0.9\pi}{2}$$

$$\frac{2\pi W}{a} = \tan\left(\frac{0.9\pi}{2}\right)$$

$$W = \frac{a}{2\pi}\tan\left(\frac{0.9\pi}{2}\right) \quad \text{in radians}$$

$$W = 1.005a = 5.025\,\text{kHz}$$

Therefore the interval or range of frequency for the BW that carries only 90% of the signal power is $[-5.025\,\text{kHz}, 5.025\,\text{kHz}]$

$$BW = 2W = \underline{\underline{10.05\,\text{kHz}}}$$

Auto Correlation refers
of correlation of the one

$$H_{xx}(f) = \int_{-\infty}^{\infty} \frac{1}{2a} e^{-a|\tau|} e^{-j2\pi f\tau} d\tau$$

For $\tau > 0$ For $\tau < 0$

$$\int_0^{\infty} \frac{1}{2a} e^{-a\tau} e^{-j2\pi f\tau} d\tau + \int_{-\infty}^{0} \frac{1}{2a} e^{a\tau} e^{-j2\pi f\tau} d\tau$$

$$\frac{1}{2a}\int_0^{\infty} e^{-\tau(a+j2\pi f)} d\tau + \frac{1}{2a}\int_{-\infty}^{0} e^{\tau(a-2j\pi f)} d\tau$$

$$\frac{1}{2a}\left[-\frac{e^{-\tau(a+j2\pi f)}}{a+j2\pi f}\right]_0^{\infty} + \frac{1}{2a}\left[\frac{e^{\tau(a-2j\pi f)}}{a-2j\pi f}\right]_{-\infty}^{0}$$

$$\frac{1}{2a}\left(-0 + \frac{e^0}{a+j2\pi f}\right) + \frac{1}{2a}\left(\frac{e^0}{a-2j\pi f} + 0\right)$$

$$\frac{1}{2a(a+j2\pi f)} + \frac{1}{2a(a-2j\pi f)}$$

$$\frac{a - \cancel{2j\pi f} + a + \cancel{j2\pi f}}{2a(a+j2\pi f)(a-2j\pi f)}$$

$$\frac{\cancel{2a}}{\cancel{2a}(a+j2\pi f)(a-j2\pi f)}$$

$$\frac{1}{a^2 - j2\pi f + j2\pi f + 4\pi^2 f^2}$$

$$= \frac{1}{a^2 + 4\pi^2 f^2}$$

Expected Revenue

(a) For Telecel

$$E[\text{Revenue}] = (0.15) \times E[T]$$

Revenue per minute

Average duration per call (in minutes)

$$E[T] = \frac{1}{\lambda}$$

$$= 0.15 \times 1/\lambda$$

$$= 0.15 \times \frac{1}{1/3}$$

$$E[\text{Revenue}] = 0.15 \times 3 = \underline{\underline{0.45 \text{ cedis}}}$$

(b) For MTN

$$E[\text{Revenue}] = (0.15) \times E[\lceil T \rceil]$$

Revenue per minute

Average rounded up duration per call

$$E[\lceil T \rceil] = \sum_{k=0}^{\infty} k \cdot \underbrace{P(k-1 < T \leq k)}_{\text{probability due to ceiling/rounding up.}}$$

$$\boxed{\text{from } E[X] = \sum_{i=1}^{\infty} p_i x_i}$$

Meaning: $P(k-1 < T \leq k)$ ⇒ if k is taken as 2, it means the actual during of the call falls between $[(2-1) < T \leq 2]$ which is $\boxed{1 < T \leq 2}$

$$E[\lceil T \rceil] = \sum_{k=1}^{\infty} k \left[(1 - e^{-\lambda k}) - (1 - e^{-\lambda(k-1)}) \right]$$

Since the call duration is a commulative exponential function $X(t) = 1 - e^{-\lambda t}$

$$E[\lceil T \rceil] = \sum_{k=1}^{\infty} k \left(e^{-\lambda(k-1)} - e^{-\lambda k} \right) = \frac{1}{1 - e^{-\lambda}}$$ a geometric sum to infinity

$$E[\lceil T \rceil] = \frac{1}{1 - e^{-1/3}} = \frac{1}{0.28347} = \underline{\underline{3.528}}$$

$$E[\text{Revenue}] = (0.15) \times (3.528) = \underline{\underline{0.529 \text{ cedis}}}$$

$$\boxed{F(\tau) = a e^{-a\tau}}$$

ⓐ Average delay, $E[F]$

$$E[F] = \int_0^\infty \tau a e^{-a\tau}\, d\tau$$ from $0 \to \infty$ because of power ≥ 0

Integration by parts $\int u\,dv = uv - \int v\,du$

Let $u = \tau$ $\quad du = 1$

$dv = a e^{-a\tau}$ $\quad v = -e^{-a\tau}$

$$\int \tau a e^{-a\tau} = \left[-\tau e^{-a\tau}\right]_0^\infty - \int_0^\infty -e^{-a\tau}$$

$$= \left[-\tau e^{-a\tau} - \frac{e^{-a\tau}}{a}\right]_0^\infty$$

$$= \left[-(\infty) e^{-a(\infty)} - \frac{e^{-a(\infty)}}{a}\right] - \left[-(0) e^{-a(0)} - \frac{e^{-a(0)}}{a}\right]$$

$$= (0) - (0 - 1/a) = \underline{\underline{\frac{1}{a}}}$$

ⓑ $\sigma_\tau^2 = E[F^2] - (E[F])^2$

$$E[F^2] = \int_0^\infty \tau^2 a e^{-a\tau}\, d\tau$$

Let $u = \tau^2$ $\quad du = 2\tau$

$dv = a e^{-a\tau}$ $\quad v = -e^{-a\tau}$

$$\int \tau^2 a e^{-a\tau} = \left[-\tau^2 e^{-a\tau}\right]_0^\infty - \underbrace{\int_0^\infty -2\tau e^{-a\tau}}$$

Another integration by parts but let's multiply by $\frac{1}{a}$ and bring the -2 out

$$= \left[-\tau^2 e^{-a\tau}\right]_0^\infty + \frac{2}{a}\underbrace{\int_0^\infty \tau a e^{-a\tau}}$$

the mean which is $\frac{1}{a}$

$$= \left[-\tau^2 e^{-a\tau}\right]_0^\infty + \frac{2}{a^2}\cancel{\int_0^\infty}$$

$$= 0 + 2/a^2 = 2/a^2$$

$$\sigma_\tau^2 = \frac{2}{a^2} - \left(\frac{1}{a}\right)^2 = \frac{2}{a^2} - \frac{1}{a^2} = \underline{\underline{\frac{1}{a^2}}}$$